Kunal Mugalkhod

Full-Stack GenAI Developer | Architecture and Technical Leadership

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PROFESSIONAL SUMMARY

Full-stack GenAI developer and System Analyst who architects applications, works with customers to define requirements, and leads GenAI project delivery. Guides the engineering team and monitors project progress while remaining hands-on with Angular, Python/FastAPI, retrieval-augmented generation (RAG), Model Context Protocol (MCP), and AWS. Creator of Lightcode, an open-source coding agent, with earlier experience in clinical-study delivery.

TECHNICAL SKILLS

Applications: Python, TypeScript, JavaScript, SQL, Angular, React, FastAPI, Flask, REST APIs

GenAI and retrieval: LangChain, LangGraph, MCP, Amazon Bedrock, Bedrock Knowledge Bases, Amazon OpenSearch, embeddings, hybrid search, Cohere reranking, LiteLLM

Cloud and delivery: AWS EKS, Docker, Kubernetes, Helm, GitLab CI/CD, S3, Lambda, API Gateway

Data and quality: PostgreSQL, DynamoDB, SQLite, Keycloak, OAuth 2.0, JWT, Langfuse, Ragas, NeMo Guardrails, CloudWatch, Pytest, JMeter

OPEN SOURCE

Lightcode | AI coding agent

github.com/kmugalkhod/lightcode | TypeScript, React, Bun, Hono, Prisma, SQLite

- Built terminal and browser interfaces sharing a local API server, agent engine, and persistent SQLite sessions; published the package as @kmugalkhod/lightcode on npm.
- Implemented permission-controlled file, shell, Git, web, and MCP tools, reusable skills, provider switching, and context management through token estimation, pruning, and summarization.

PROFESSIONAL EXPERIENCE

Cybage Software Pvt. Ltd. Aug 2021 - Present

System Analyst | Pune

- Work directly with customers to understand business needs and translate requirements into GenAI solution designs and implementation plans.
- Architect GenAI applications across frontend, backend, agent workflows, retrieval, authentication, and cloud deployment.
- Lead GenAI project delivery, monitor progress, and provide technical direction and guidance to the team through implementation, testing, and deployment.

Release Assistant | May 2025 - Present

- Built Angular/FastAPI workflows and Jira/GitLab MCP servers for release notes, environment-tag comparisons, sprint analytics, security reports, and XRAY test-execution analysis.
- Reduced candidate tickets for commit analysis from about 1,000 to 300 (approximately 70%) by filtering for tickets with commits before downstream processing.
- Integrated Amazon Bedrock and LiteLLM with Keycloak authentication and AWS EKS deployment; generated Word/CSV reports with source links and S3 downloads.

LTSA AI Assistant and LT Examination | Jan 2024 - Apr 2025

- Developed enterprise RAG assistants using Amazon OpenSearch and Bedrock Knowledge Bases, with ingestion, chunking, embeddings, hybrid retrieval, metadata filters, and Cohere reranking.
- Implemented conversational context and NeMo Guardrails; evaluated responses with Ragas and instrumented LLM workflows with Langfuse for trace review.

PROFESSIONAL EXPERIENCE CONTINUED

Cybage Software Pvt. Ltd. | System Analyst

Code Atlas | RAG for developer onboarding

- Built a code-ingestion pipeline that brings GitHub source code into a vector database for retrieval across unfamiliar repositories.
- Connected code retrieval with LLM response generation to answer implementation questions using relevant repository content as context.
- Enabled natural-language code exploration for new team members, supporting developer onboarding and understanding of existing implementations.

BPMN 3.0 | Agent workflow project

- Built separate agents for diagram and edge generation, followed by validation of the generated outputs.
- Regenerated invalid outputs and validated them again before merging the accepted diagram and edges into the final workflow.

LiteLLM Gateway and SSO Portal

- Deployed a LiteLLM gateway on AWS EKS for Bedrock models and integrated Release Assistant and Claude Code through a shared model-access layer.
- Built an Angular/FastAPI portal with Keycloak SSO, per-user and team virtual keys, budgets, and model-level usage and cost visibility backed by PostgreSQL.

API Catalog and OpenAPI Governance

- Built an API Catalog POC and reusable GitLab pipeline for versioned documentation publishing to S3 on release tags; integrated 20 Spectral rules and a dedicated validation stage to surface OpenAPI specification issues.

Earlier engineering projects at Cybage

Built Python REST APIs and PostgreSQL workflows using AWS Lambda, Step Functions, and API Gateway; developed Amazon Textract extraction for land-title data. Earlier ML work includes image classification and ARIMA, SARIMA, and Prophet forecasting with MAE/RMSE evaluation.

EARLIER PROFESSIONAL EXPERIENCE

Oracle Solution Mar 2019 - Jul 2021

Senior Application Consultant | Pune

- Led clinical-study delivery for life sciences projects, including PPC activities and coordination of study setup and deployment.
- Reviewed and validated study documentation and managed study deployment to production, ensuring the deployed study matched the approved requirements.

Veeva Clinical Research Jul 2018 - Feb 2019

Associate Data Scientist | Ahmedabad

- Managed clinical-study setup and delivery for life sciences projects, supporting the collection and processing of clinical-trial participant data.
- Reviewed study documentation, validated study configurations, and deployed clinical studies to production.

Syneos Health / inVentiv Health Feb 2013 - Jun 2018

Senior Clinical Data Programmer | Pune

- Built and configured clinical studies using Oracle InForm and Central Designer, translating study requirements into study configurations and data-collection workflows.
- Reviewed and validated study documentation and configurations, supported study management, and deployed clinical studies to production.

EDUCATION AND CERTIFICATION

Bachelor of Engineering | Shivaji University | 2011

C-DAC Professional Course | 2012 | Base SAS Certification